



Massive Martian Storm

A dust storm has engulfed the red planet, and it looks like the storm will continue to rage till at least September. <https://digit.in/Aug18-60-1>



Martian Climate Study

Researchers are studying the minerals on Mars to figure out the climate on Mars thousands of years ago. <https://digit.in/Aug18-60-2>

MANGALYAAN: STILL ALIVE

Four years after demonstrating ISRO's capability for an interplanetary mission, Mangalyaan continues to beam back data and shows no sign of stopping

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y the time you are reading this, India's first mission to Mars, Mangalyaan will have completed over 1400

days, or 46 months, in orbit around Mars. The original planned mission duration was only 6 months. The orbiter was supposed to be merely a technology demonstration mission, to give the Indian researchers first hand experience of an interplanetary mission. The scientific objectives were secondary. However, the orbiter continues to observe Mars and gather up to date data, which is released to the scientific community. We spoke to Spacecraft Controller of Mars Orbiter Mission (MoM), Kamallesh Kumar Sharma about the spacecraft, its future prospects, and the science benefits.

One of the first questions we asked was, how long could the mission extend, and what are the science objectives? Sharma replied, "since the

satellite is in very good condition and still there is a good amount of fuel left, the mission may extend to many more years. Current and future objectives are basically to analyse the science data of the different instruments."

There are five scientific instruments on board the spacecraft. The Methane Sensor for Mars (MSM) uses reflected sunlight to measure methane in the atmosphere with an accuracy of particles per billion. The instrument is also very good at just measuring the reflected sunlight from Mars, a property known as its albedo. The Mars Colour Camera (MCC) captures the images of the Martian surface, can monitor the red planet for dynamic weather events, and provides a context for the data gathered by the other sensors on board. The Lyman Alpha Photometer (LAP) is used to measure the amount of deuterium and hydrogen in the upper atmosphere. The mineralogical studies of the surface of the planet is done through the Thermal Infrared Imaging Spectrometer (TIS) instrument. The instrument measures the

thermal emissions on the surface, and the types of soil or minerals can be identified based on the thermal characteristics observed. The instruments are all healthy and working as expected. Mars Exospheric Neutral Composition Analyser (MENCA) measures the neutral gases in the atmosphere, such as nitrogen, oxygen and carbon dioxide. The instrument was developed by the Vikram Sarabhai Space Centre (VSSC).

One of the surprising findings by the MENCA instrument was the discovery of superthermal or hot argon in the upper atmosphere, or exosphere of Mars. The particles were discovered at unexpected altitudes. The finding could explain the process by which the atmosphere of Mars has been stripped away. The Martian atmosphere is being stripped away constantly, because of solar activity. This is just one of the many aspects of Mars that the instruments on board MoM continue to study. "Science payloads analysis is underway including possibility of methane, if it exists, how the atmosphere of Mars disappeared, what are the neutral components of the Martian atmosphere, thermals mapping, mineralogy mapping etc," explains Sharma.